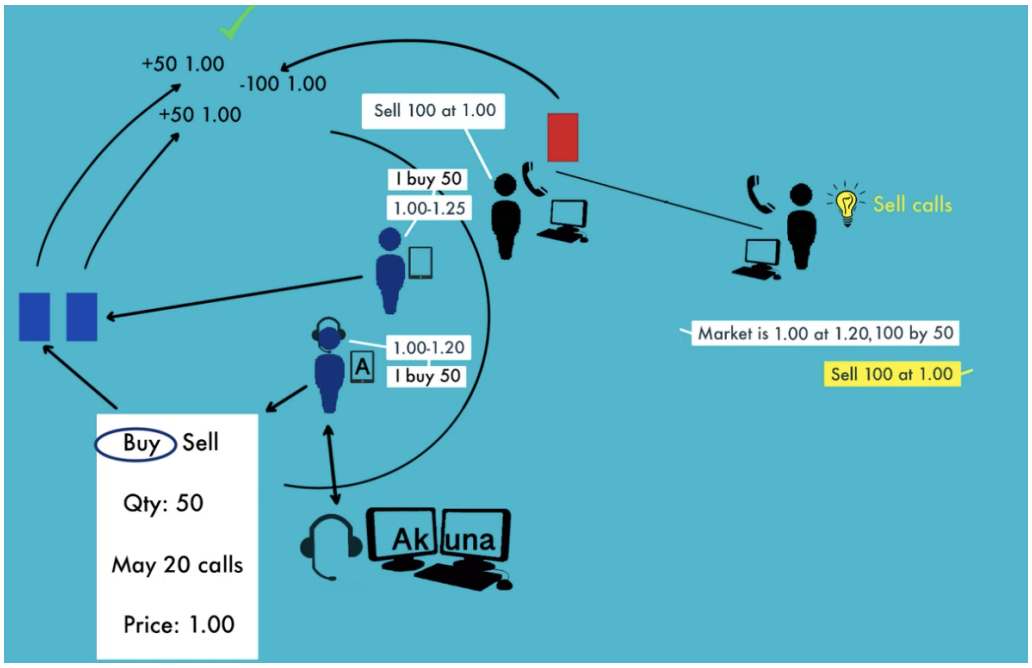

Figure 4: Example of an off-floor seller requesting a market and then selling into the best bid quoted by the crowd.

8.6 Floor Trading Versus Electronic Trading

Remark 8.6. On an electronic exchange, the same process is more automated. Quotes are displayed centrally, orders are entered directly into the exchange, fills are allocated electronically, and clearing systems receive the trade data almost immediately.

Remark 8.7. Floor trading relies more heavily on human communication and manual confirmation, while electronic trading reduces human error and speeds up execution and reporting. Even so, floor information can still matter, which is why some firms stay active in both venues.

9 Forwards, Futures, and Options on Futures

9.1 Why Forward Contracts Exist

Definition 9.1 (Forward Contract)**.** A **forward contract** is an agreement between two parties to transact an asset at a specified future date for a specified price.

Remark 9.1. A forward contract is designed to reduce uncertainty. Instead of waiting for the market price at harvest or delivery time, both sides agree in advance on the price at which they will trade.

Example 9.1 (Apple Farmer and Pie Chain)**.** Suppose an apple farmer expects to produce 1 million pounds of apples this year. The farmer faces price risk: if apple prices fall too much by harvest time, the farmer may not even cover costs. On the other side, a pie chain needs apples as an input and faces the opposite risk: if apple prices rise too much, its margins may collapse.

To reduce this uncertainty, the farmer and the pie chain can agree today that at harvest time the pie chain will buy 1 million pounds of apples for \$0.20 per pound.

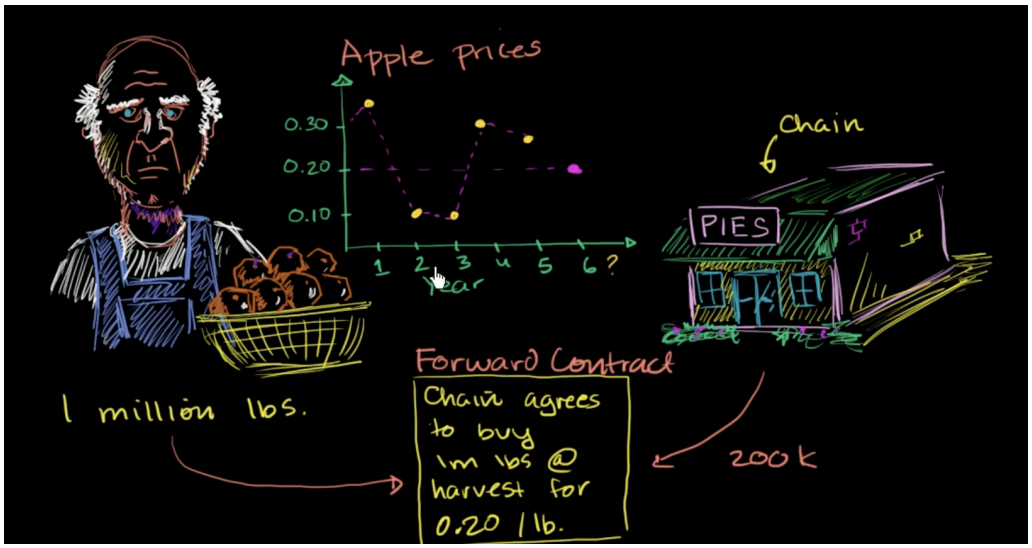

Figure 5: A forward contract lets a producer and a consumer lock in a future transaction price, reducing uncertainty for both sides.

Remark 9.2. This agreement benefits both sides. The farmer gains predictability and can plan around a known selling price. The pie chain gains cost certainty and avoids being exposed to large upward price moves in apples.

9.2 Obligation in a Forward Contract

Remark 9.3. A forward is not merely an option to transact later. It is a bilateral obligation. On the delivery date, the seller must deliver the asset and the buyer must provide payment at the agreed price.

Example 9.2. If the agreed price is \$0.20 per pound for 1 million pounds of apples, then on the delivery date:

- the farmer must deliver 1 million pounds of apples, and
- the pie chain must pay \$200,000.

9.3 From Forwards to Futures

Definition 9.2 (Futures Contract)**.** A **futures contract** is a standardized exchange-traded agreement to buy or sell an asset at a specified future time and price.

Remark 9.4. Futures are closely related to forwards, but unlike forwards they trade on an exchange and use standardized contract terms. This standardization makes them easier to trade, offset, and clear.

Remark 9.5. The exchange stands between the buyer and seller. This is what makes it possible for one participant to buy from one party and later sell to another without being left with a messy custom agreement.

9.4 How the Exchange Makes the Market Work

Remark 9.6. An exchange can profit by standing in the middle of buyers and sellers. For example, it may effectively buy exposure from one side at one price and sell exposure to the other side at a slightly better price, earning the spread.

Example 9.3 (Exchange Spread)**.** Suppose the exchange facilitates contracts where one side effectively sells at \$0.20 per pound and the other side buys at \$0.22 per pound. The exchange earns a spread of

$$0.22 - 0.20 = 0.02$$

per pound.

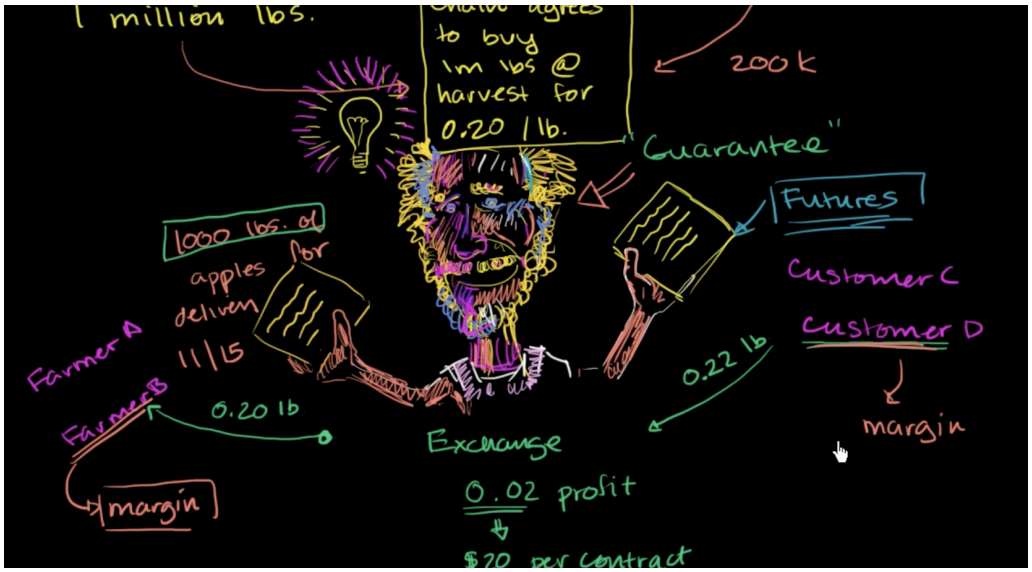

Figure 6: The exchange stands in the middle of the transaction, earning the spread while requiring margin from participants to reduce default risk.

Remark 9.7. If the contract covers 1,000 pounds, then a \$0.02 spread corresponds to

$$1000 \times 0.02 = 20$$

dollars per contract.

9.5 Margin and Counterparty Protection

Definition 9.3 (Margin)**. Margin** is money that market participants must post as a financial cushion against adverse price movement.

Remark 9.8. Margin helps the exchange manage counterparty risk. If prices move sharply against one side of a contract, the posted margin provides protection and helps ensure that both parties can still meet their obligations.

Remark 9.9. This is one of the central differences between exchange-traded contracts and purely private agreements. Exchange markets reduce the risk that one side simply fails to perform.

9.6 Producers and Consumers

Definition 9.4 (Producer and Consumer Hedging)**.** A **producer** is a participant who expects to own or create the underlying asset in the future, while a **consumer** expects to need that asset in the future.

Remark 9.10. Producers and consumers often use futures to lock in prices. The producer protects against falling prices, while the consumer protects against rising prices.

Example 9.4. A farmer expecting to harvest wheat in November may sell November wheat futures now. If wheat prices fall later, the farmer loses in the cash market but gains on the futures position. If wheat prices rise later, the farmer gains in the cash market but loses on the futures position. In either case, much of the price risk has been hedged.

9.7 Options on Futures

Definition 9.5 (Option on a Future)**.** An **option on a future** is an option whose underlying instrument is a futures contract rather than a stock or a cash index.

Remark 9.11. This means the option does not directly turn into the physical commodity. Instead, if exercised, it creates or transfers a futures position at the strike price.

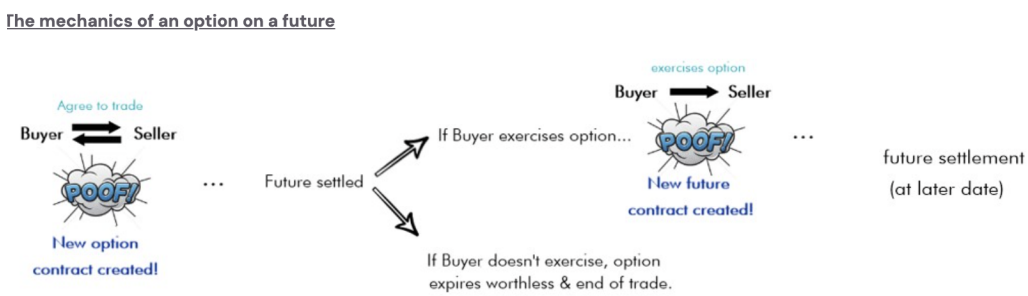

Figure 7: Mechanically, an option on a future either expires worthless or turns into a futures position if exercised. The resulting future then follows its own settlement process later.

9.8 Mechanics of Exercise

Example 9.5 (Exercise Outcome)**.** Suppose Trader A buys a call option on a future from Trader B. At expiry, if the option is worth exercising, Trader A exercises it and receives a futures position at the strike price. Trader B, as the seller, must deliver the corresponding futures position.

Remark 9.12. After exercise, the option contract is gone, but the futures position remains. That futures position may later be closed out, cash settled, or physically delivered depending on the contract.

Remark 9.13. If the option is not worth exercising at expiry, then it simply expires worthless and no futures contract is created from that option.